

Formation Control of Multi-UAVs Using Negative-Imaginary Systems Theory

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Abstract— We present a new formation and consensus control architecture for a team of multiple Unmanned Aerial Vehicles (UAVs) using negative-imaginary (NI) systems theory performing cooperative tasks such as rescue missions, disaster area exploration or agriculture. A linear velocity control algorithm for the lateral and longitudinal directions is proposed to stabilize each UAV and to meet the requirements of a NI plant. After that, a formation for UAVs is selected and used for providing reference points. Then, a collaborative structure which enables the UAVs to follow the predefined formation is illustrated. Finally, some typical examples of formation control for two UAVs on a plane are implemented in simulation and in real-flight to validate the efficiency of our proposed control architecture.

I. INTRODUCTION

NI systems theory has drawn much attention in recent years [1]–[3], since it was first fully presented in [1]. This is because there is a wide class of successful applications of NI systems theory, which arises in many high technology applications such as aerospace, nano-positioning control [4], multi-DOF robotic arms and the vibration control of flexible structures with high amplitude oscillation [5]. In these systems, the input is a force and the output is a position.

The definitions of strictly-negative-imaginary (SNI) systems for a single-input single-output (SISO) or multi-input multi-output (MIMO) system and their consensus theory can be found in [2], [6], [7]. However, these robust cooperative control theories have never been applied to consensus and formation control for multi-UAV systems although this topic has been an active research area recently [8]–[12]. Many state-of-the-art control laws in [13], [14], which are successfully implemented on one UAV, are unable to be extended to control multiple UAVs or are not well combined with network protocols. Other papers have paid attention only to a formation methodology on planar, but the closed-loop control stability for each UAVs on two axes was not sufficiently addressed; e.g. [12], [15], [16]. This deficiency can cause asynchronism between position/velocity control consensus algorithm, resulting in imperfect cooperative performance.

Additionally, agents considered in [2], [16] for formation and consensus control are treated as second order systems. Similarity, in [12], the outer loops of position/velocity control of each UAVs are second order. However, a closed-loop position/velocity control of one UAV must be at least fourth order.

According to the consensus theory in [2], the UAV leader usually performs its tasks relating to consensus and movement at the same time. Nevertheless, this method is unable to be implemented in practice since the UAV leader must wait for each slave UAV to reach its current consensus point, before moving to a new point in the configured flight path. In the consensus scenario for multi-UAVs, instead of being moved to the same point in space, each UAV should be simultaneously driven to its own unique desired positions in the formation. Next, the equations for consensus between the master UAV and the slave UAVs in [2] are asymmetrical. This causes practical difficulties in tuning the gain values of the SNI controllers.

In an effort to improve the formation control of multi-UAVs, this paper addresses the existing problems of formation and consensus control mentioned above using negative-imaginary (NI) systems theory. In our work, a velocity control algorithm will be selected so that it is not only able to stabilize each UAV's velocity in the longitudinal (x) and lateral (y) axes but also to satisfy the criteria of an NI closed loop system. From there, according to the NI systems theory [2], if these NI systems are controlled by a SNI controller, robust stability for the whole network will be achieved. Moreover, a new structure of consensus control with improved network properties is proposed in such a way that each follower UAV will travel and build a consensus between its relative position and that of the leader UAV following a predefined formation on a plane. In outdoor scenarios, we assume that the master UAV is equipped with a GPS to determine its absolute position. However, we further assume that the slave UAVs are only fed precise relative positioning with respect to the master UAV (not an absolute position) which is measured by sensors such as a machine vision system or a radio beacon. In our indoor experiments, these data are simulated and measured using a Vicon Motion Capture System. Finally, some typical examples of formation control for two UAVs on a plane are conducted in simulation and in real flights to validate this new theory.

II. EXPERIMENTAL APPARATUS

The following software and hardware have been used to conduct the experiments:

1) UAVs: F450 quadrotor platforms (as in Fig. 1) were built so as to collect the necessary data of orientation and position.

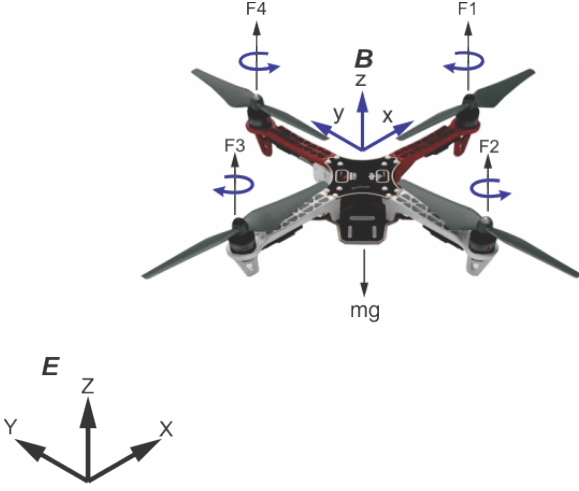


Fig. 1. A diagram of the DJI F450 quadrotor

2) *Wifi Network*: This network is used to transmit data at frequency of 100 Hz. The network provides communication between a notebook or ground station (GS) and the quadrotors.

3) *Vicon Motion Capture System*: is used in combination with reflective markers attached to each quadrotor for analyzing the motion of quadrotors, including actual position and orientation. The data of each UAV will then be communicated to the GS and each quadrotor via the Wifi system code.

4) *Robot Operating System (ROS) and MAVROS*: An open source framework for communication between the robots and the GS, is selected to facilitate information exchange as well as to control the quadrotors through the services provided.

III. BACKGROUNDS

A. The Basics of Graph Theory

1) *Information Graph*: $G_i = (V, E)$ is defined as an information graph of n nodes and l edges. In there, V contains the set of nodes $(V_1, V_2, \dots, V_n)$ and $E \subseteq (V_i, V_j)$ are a set of the ordered pairs of edges between two nodes in V .

B. The Information Direction Matrix

The matrix Q_a is a $(n \times l)$ matrix, whose edges have been arbitrarily oriented and described as

$$Q_a = \begin{cases} V_{ij} = 1 & \text{if } V_i \text{ is the head of } E_{V_i, V_j} \\ V_{ij} = -1 & \text{if } V_i \text{ is the tail of } E_{V_i, V_j} \\ V_{ij} = 0 & \text{if } V_i \text{ is not connected to } V_j \end{cases} \quad (1)$$

1) *The Consensus Matrix*: A consensus matrix Q_c ($n \times l$) describes whether a node or UAV will communicate with another UAV in the group or not.

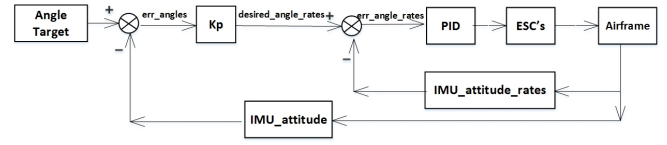


Fig. 2. A cascaded PID structure for controlling the attitude of a quadrotor

$$Q_b = \begin{cases} B_{ij} = 0 & \text{if } UAV_i \text{ is the head of } E_{V_i, V_j} \\ & \text{for consensus} \\ B_{ij} = -1 & \text{if } UAV_i \text{ is the tail of } E_{V_i, V_j} \\ & \text{for consensus} \\ B_{ij} = 0 & \text{otherwise} \end{cases} \quad (2)$$

2) *The Reference Matrix*: A reference matrix Q_r ($n \times 1$) = $\text{diag}_{i=1}^n R_i$ presents which node or UAV in group will refer to the rendezvous point

$$Q_r = \begin{cases} R_i = 1 & \text{if } UAV_i \text{ is connected to the reference} \\ R_i = 0 & \text{if } UAV_i \text{ is not connected to the reference} \end{cases} \quad (3)$$

C. The Cooperative Control Theory for NI Systems

A transfer function $P(s)$ of a system whose Nyquist plot completely lies below the real axis for all positive frequencies ($j[P(j\omega) - P^*(j\omega)] > 0$ for all $\omega > 0$), is called a SNI system. Similarly, a transfer function $P(s)$ is NI if its Nyquist plot for $\omega \geq 0$ is contained in the lower half of the complex plane, but does not cross the real axis ($j[P(j\omega) - P^*(j\omega)] \geq 0$ for all $\omega \geq 0$). In [1], an NI transfer function is able to be robustly controlled by an SNI controller in a positive feedback loop. This fundamental principle has been used recently to develop consensus control approaches for NI systems. If a multi-input multi-output (MIMO) system or multi-agent system contains a group of NI transfer functions, connected with their SNI controllers via a cooperative information graph, these control systems are guaranteed to have robustness and stability. The main result in [2] is that all NI agent outputs, under the control of SNI controllers, will reach their assembly point from different initial positions.

IV. METHODOLOGY

A. Dynamic Modeling of Velocity Control

For convenience, a PID control loop was designed using the quadrotor's roll, pitch and yaw angles in order to stabilize the quadrotor during the take-off process. The overall diagrams of the attitude control system are illustrated in Fig.2.

Once all relevant data from the attitude control system are stored in a text file after flight tests, it is analyzed using a MATLAB script to identify the transfer function of the plant between actual angles (*cur-roll/cur-pitch*) and angular set points (*roll-sp/pitch-sp*) via an ARMAX model (Tab. I). The overall equations for this model are described in z domain as follows

$$A(z) * y(k) = B(z) * u(k - n) + e(k) \quad (4)$$

TABLE I

THE TRANSFER FUNCTION BETWEEN ROLL/PITCH SET POINT AND ACTUAL ROLL/PITCH ANGLE

	Transfer function
Roll angle	$\text{cur-roll/roll-sp} = \frac{-0.086369s+17.6425}{s^2+4.6787s+27.3284}$
Pitch angle	$\text{cur-pitch/pitch-sp} = \frac{-0.28504s+3.0135}{s^2+5.2064s+11.258}$

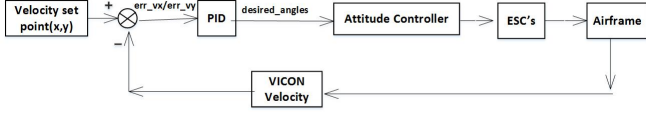


Fig. 3. A PID structure for controlling the velocity of a quadrotor

where $u(k)$ is the system input, $y(k)$ is the system output, n is the system delay, k is the present time and $e(k)$ is the disturbance in system.

Furthermore, this attitude control loop is treated as an inner loop, and an added velocity control loop using a PID controller (Fig.3) is used as an outer loop. The parameters of these PID controllers on planar plane are then tuned properly so that the lateral and the longitudinal velocity control channel for planar is SNI or NI. The proposed velocity control structure (Fig.3) is applied to the UAV during experiments and its velocity data ($Vel_x, Vely$) are collected at the same time. Later, the velocity transfer functions between the velocity set point input ($velx-sp/vely-sp$) and the current velocity output ($curr-velx/curr-vely$) for the x and y axis (Tab. II) are found by the same model as the roll and pitch transfer functions. As seen from their Nyquist plots (Fig.4), these linear velocity transfer functions are SNI. To sum up, using NI systems theory, it is observed that an UAV with a fourth order SNI velocity transfer function can be robustly stabilized on a horizontal plane using a SNI controller.

B. Consensus and Formation Control Structure for Multi-UAVs

As presented in [2] and [6], the transfer function of agent $i \in (1..n \text{ agents})$ in a multiple NI system with $n > 1$ is (Fig. 5)

$$y_i = P_i(s) * u_i \quad (5)$$

where $y_i \in \mathbb{R}^{m_i \times 1}$ and $u_i \in \mathbb{R}^{m_i \times 1}$ are the input and output of the transfer function of agent i respectively. $P_i(s)$ is the SNI/NI transfer function of agent i . In multi-UAVs applications, $P_i(s)$ is the SNI/NI velocity transfer function of the UAV_i on a horizontal plane.

This NI system plant is then connected with the matrix of reference, direction and consensus matrices (Q_a, Q_b, Q_r) to

TABLE II

THE TRANSFER FUNCTION BETWEEN VELOCITY X/VELOCITY Y SET POINT AND ACTUAL VELOCITY X/ VELOCITY Y

	Transfer function
Velx	$\text{curr-velx/velx-sp} = \frac{-0.0437s^3+5.16s^2-1860.11s+54489.05}{s^4+86.155s^3+54490.53s^2+29510.63s+1481.6}$
Vely	$\text{curr-vely/vely-sp} = \frac{-0.0436s^3+4.24s^2-1494.74s+48347.83}{s^4+88.9s^3+54883.91s^2+49242.03s+1266.25}$

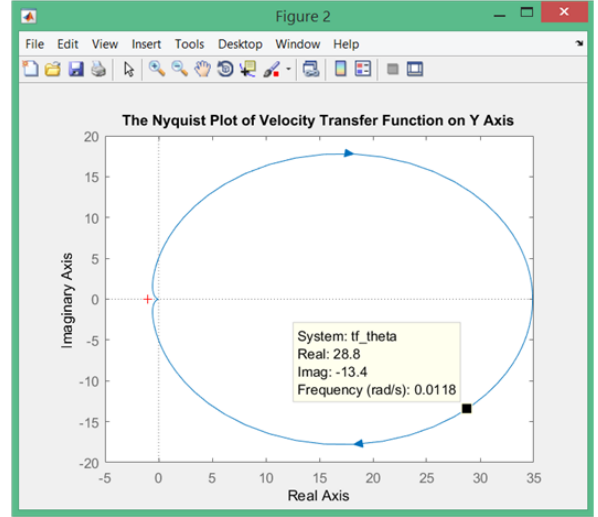
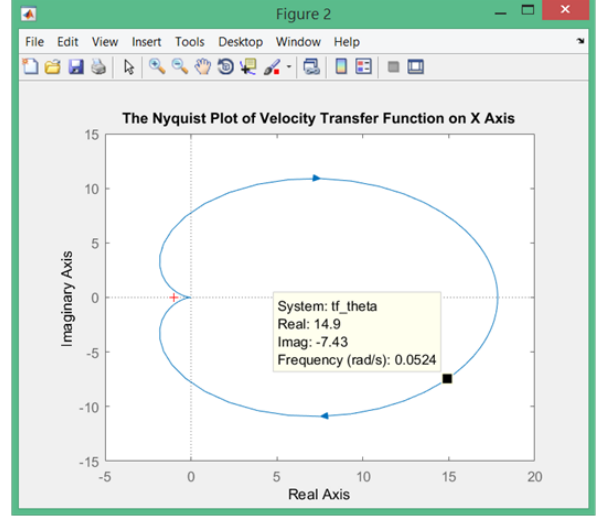


Fig. 4. The Nyquist plots of the velocity transfer functions for the x and y axes.

generate the overall network plant which is depicted in Fig. 6. Thanks to the new communication topology structure, any UAV which implements the formation and consensus, will automatically receive two significant pieces of information. An UAV can be either the master UAV or a slave UAV [1], and it should move to the relative position or the rendezvous point [2].

Regarding the consensus output feedback law [2] and [6], when an agreement is achieved, all of the agents will be driven from their initial positions to the final rendezvous point. Nevertheless, this approach is unable to be applied in multi-UAV control as explained in the introduction. It is evident that offset distances between the UAVs in the lateral and longitudinal axes, produced from the pre-defined formation, are essential. To resolve this problem, we now define a full consensus control structure that considers the formation, reference point and consensus factors (Fig. 7).

The overall equations for the new structure are as follows

$$X_r = 1_n \otimes r \quad (6)$$

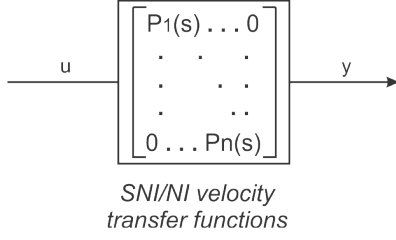


Fig. 5. Multiple NI plants

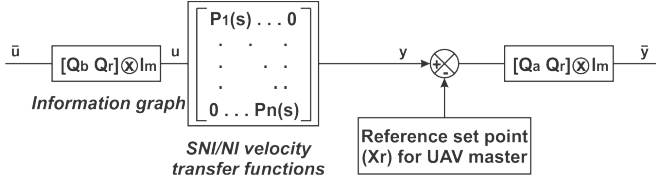


Fig. 6. Overall closed-loop network plant

$$e_r = y_i - X_r \quad (7)$$

$$\bar{y} = ([Q_a \ Q_r] \otimes I_m) e_i \quad (8)$$

$$e_f = \bar{y} - X_f \quad (9)$$

$$u = ([Q_b \ Q_r] \otimes I_m) \text{diag}_{i=1}^n P_s(s) e_f \quad (10)$$

where $\otimes$ is the Kronecker product, the input of this plant $u \in \mathbb{R}^{m \times 1}$ is the velocity set point of each UAV, the output $y \in \mathbb{R}^{m \times 1}$ is the current position of each UAV. X_r is the reference set point of the master UAV and X_f is the relative position between the slave UAVs and the master UAV in the configured formation. $\bar{u} \in \mathbb{R}^{lm \times 1}$ and $\bar{y} \in \mathbb{R}^{lm \times 1}$ are the input and output of overall network plant. $P_s(s) \in \mathbb{R}^{lm \times lm}$ is the group of SNI consensus and tracking controllers for the multiple UAVs.

The multi-UAVs under the control of SNI controllers and the new information network topology are guaranteed to converge to the rendezvous point while maintaining a desired spatial pattern.

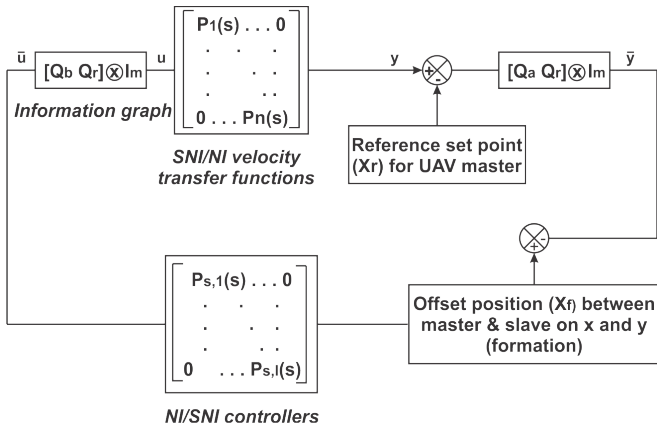


Fig. 7. The consensus and formation control structure for SNI multi-UAVs through the information graph.

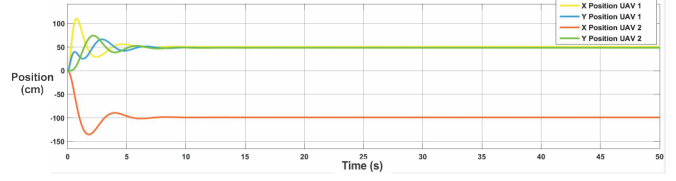


Fig. 8. Case 1: Position response of the two UAVs

V. SIMULATION RESULTS

Three cases will be given to illustrate the main results of this paper. A master UAV will quickly move to the position set point with known X_r , while a slave UAV will coordinate with the master UAV via an offset distance defined from X_f and master UAV's position.

The main set of equations describing the consensus and formation control for two UAVs using NI systems theory is summarized as follows

$$u = \begin{bmatrix} Vxsp_1 \\ Vysp_1 \\ Vxsp_2 \\ Vysp_2 \end{bmatrix} = \begin{bmatrix} Kr_x^*(X_1 - X_{r,ref}) \\ Kr_y^*(Y_1 - Y_{r,ref}) \\ -Kc_x^*(X_1 - X_2 - X_f) \\ -Kc_y^*(Y_1 - Y_2 - Y_f) \end{bmatrix} \quad (11)$$

$$= \begin{bmatrix} Kr_x^*(X_1 - X_{r,ref}) \\ Kr_y^*(Y_1 - Y_{r,ref}) \\ -Kc_x^*(X_l - X_f) \\ -Kc_y^*(Y_l - Y_f) \end{bmatrix} \quad (12)$$

where Kr_x , Kr_y and Kc_x , Kc_y are the reference and consensus gain values of the NI controllers on the two axes x and y. $(Vxsp_1, Vysp_1)$, (X_1, Y_1) and $(Vxsp_2, Vysp_2)$, (X_2, Y_2) are the absolute position and linear velocity of the master UAV and the slave UAV on the horizontal plane. X_l and Y_l are the relative position between the leader and the follower on the x and y axes.

The output equation of the SNI plant is:

$$y = \begin{bmatrix} X_1 \\ Y_1 \\ X_2 \\ Y_2 \end{bmatrix} = P(s) * u \quad (13)$$

where $P(s)$ is the SNI velocity transfer function matrix of the two UAVs on x and y axes.

Case 1: the NI controllers are the constant gains $(-1.2, -3.6, -9, -8.95)$, $X_r=(50, 50)$ cm, $X_f=(150, 0)$ cm, $Q_a = [1; -1]$, $Q_b = [0; -1]$, $Q_r = [1; 0]$. It can be seen from (Fig. 8) that robust formation control is guaranteed via the consensus and formation control law for multi-UAVs. UAV_1 (master UAV) is driven to the rendezvous point (50, 50) cm after 5 seconds while UAV_2 (slave UAV) is stabilized in its position at the point (-100, 50) cm because the offset position is (150, 0) cm (Fig. 9).

Case 2: the NI controllers are the constant gains $(-1.2, -3.6, -9, -8.95)$, $X_r=(50, 50)$ cm, $X_f=(150, 50)$ cm, $Q_a = [1; -1]$, $Q_b = [0; -1]$, $Q_r = [1; 0]$. Similarly, UAV_1 reaches the reference point (50, 50) cm after 5 seconds while UAV_2

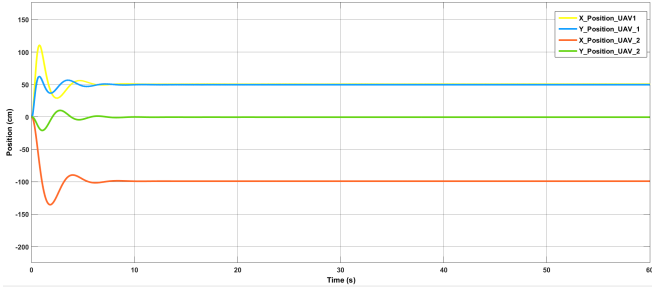


Fig. 9. Case 2: Position response of the two UAVs

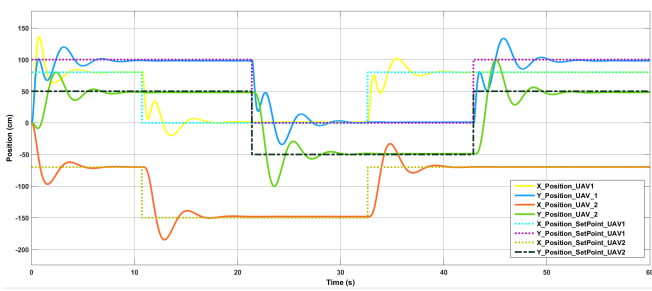


Fig. 10. Case 3: Position response of the two UAVs

moves to the point (-100, 0) cm because the offset position is (150, 50) cm (Fig. 9).

Case 3: the NI controllers are the constant gains (-1.2, -3.6, -9, -8.95), $X_f = (150, 50)$ cm, $Q_a = [1; -1]$, $Q_b = [0; -1]$, $Q_r = [1; 0]$. The X_r parameter will be assigned to the four rectangular coordinates, including [80, 100], [0, 100], [0, 0], [80, 0]. Based on the consensus structure, the master UAV must travel around these four given points from the starting point at [0, 0] and the slave UAV follows to its four point targets [-70, 50], [-150, 50], [-150, -50], [-70, -50] from the same initial point. Each UAV will wait for each other until the two go into their defined rectangular vertexes and then hover at each point for 1.5 seconds. According to Fig. 10 and Fig. 11, it is evident that UAV_1 and UAV_2 perform a strong consensus under the control of NI controllers and the communication graph.

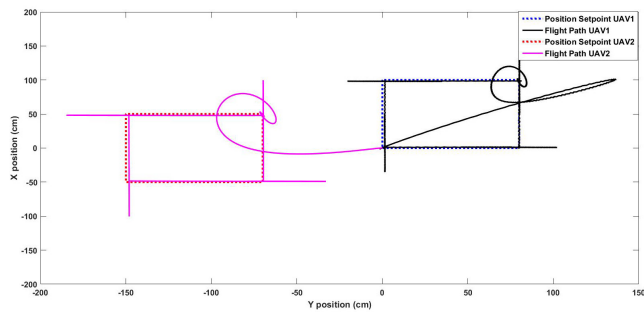


Fig. 11. The simulation results for a rectangular formation of the two UAVs.

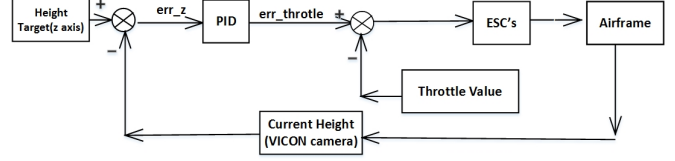


Fig. 12. The altitude controller for the two UAVs using PID algorithm

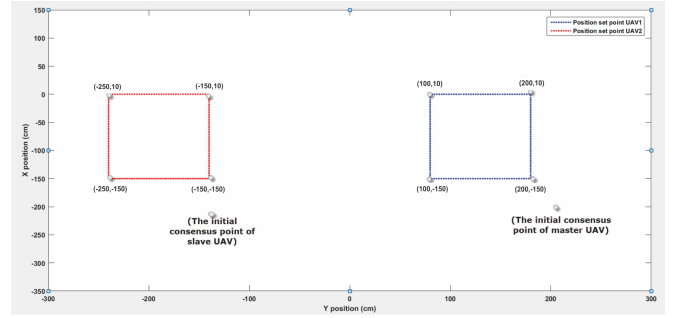


Fig. 13. The position set points of the two UAVs on a plane.

VI. EXPERIMENTAL FLIGHT TESTS

Firstly, a PID altitude and yaw controller is used to maintain the height of two UAVs on the z axis at 80 cm as well as to fix their headings at 90 degrees before the experiments for the consensus and formation control of two UAVs are able to be conducted (Fig. 12). After that, a motion planner is designed to drive the two UAVs following a rectangular formation. The master UAV will be referred to this flight planner for identifying the next position set point on a plane while the slave UAV will travel following the flight path of the master UAV. In addition, at each consensus point, the two UAVs must wait for each other until the two reach their own reference points. The location of reference points (X_r) for the master UAV is [the initial point of consensus on x axis, the initial point of consensus on y axis], [200, -150], [200, 10], [100, 10], [100, -150] cm. Based on these reference points for the master UAV, the consensus points for the slave UAV should be [the initial point of consensus on the x axis; the initial point of consensus on the y axis], [-150; -150], [-150; 10], [-250; 10], [-250; -150] (Fig. 13) if the (X_f) is (350, 0) cm. The constant NI gain values for the master UAV and the slave UAV are (-0.7, -0.7) and (-0.7, -0.7), respectively. While the two UAVs return to the initial rectangular vertex ([200, -150] for the master UAV and [-150; -150] for the slave UAV), the two will automatically land by themselves at these points.

As seen in Fig. 14, the master is able to reach the five different points on a horizontal plane. Additionally, consensus and an offset distance from the position of the master to that of the slave are guaranteed during flight test. The flight paths of the slave on the x and y axes are similar to those of the master.

VII. CONCLUSIONS AND FUTURE WORK

In the dedicated collaborative strategy using NI systems theory, the formation and reference information is integrated

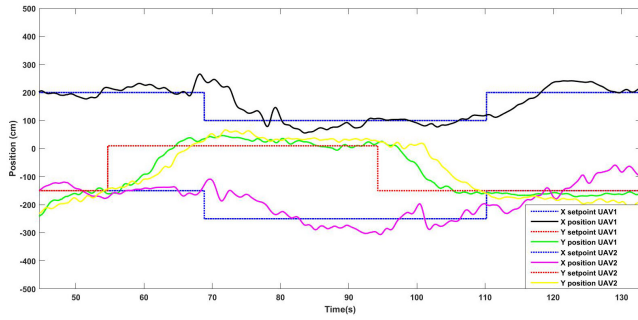


Fig. 14. Position response of the two UAVs in the real flight test

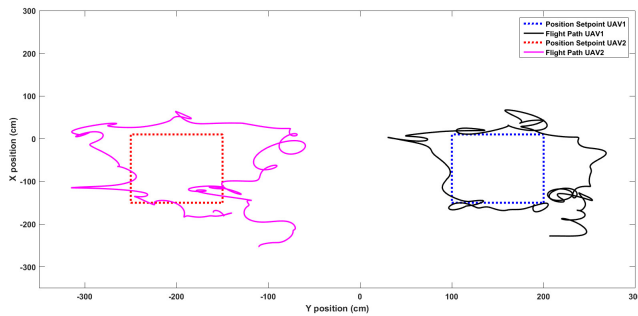


Fig. 15. The rectangular formation of the two UAVs

in a communication graph that enables all of the UAVs to follow a pre-defined trajectory.

According to Fig. 14, the slave UAV is quite sensitive to the sudden alterations in position of the master UAV. Consequently, the slave UAV makes a slightly bigger position change. Moreover, Fig. 15 shows that there are some twists at each rectangular vertex. The main reason is that any UAVs staying at one of the five reference points must wait for the consensus of the remaining UAVs at their own references by hovering around that point. This is essential because there is a short delay in the position response of the slave UAV as a result of alteration in master UAV's position. This also leads to an error in starting position while the two UAVs begin to reach the next point. Furthermore, the air flow generated between two UAVs in the flight test in case of participation of the two, also significantly affects the tracking performance. The final problem encountered is the latency in data analysis and publishing.

The accuracy of the trajectory of the leader and the follower is able to be dramatically improved in future by using a linear velocity sensor attached to each UAV. By this way, the velocity data on planar of UAVs are collected better.

Our indoor facility is not large enough to allow a larger rectangular trajectory of desired motion for the leader and the follower. Hence, the close loop tracking error of the drone is not that different to the scale of the rectangle.

In future work, the formation of multi-UAVs using the NI systems theory will be tested further. For the master-slave strategy, a master, which is a ground vehicle, and two-UAVs in a group will be driven following a certain pattern in space such as a rectangular or triangular trajectory.

Furthermore, this consensus control structure is also able to be applied in leaderless situations or flocking behaviors such as cohesion, separation or obstacle avoidance since the relative reference positions and angle set points for each agents are derived from the pre-described communication graph. This interesting topic will be researched in next papers.

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